

Gym Project – Detailed Explanation of models.py

1. Why Django Choices Have Two Values

In Django models, choice fields are defined using tuples containing two values such as ('chest', 'Chest'). The first value is the actual value stored in the database, while the second value is the human-readable label shown in forms and the Django admin panel. This separation ensures database consistency and avoids issues related to case sensitivity or formatting. When a user selects 'Chest' in the admin panel, Django stores 'chest' internally in the database.

2. Execution Order: Class Body vs `__str__()` Method

When Django loads a model, Python executes the class body first, registering fields and methods. The `__str__()` method is not executed during class loading. It is only executed when an instance of the model is converted into a string, such as when Django renders objects in the admin panel using `str(object)`.

3. Multiple `__str__()` Methods and Python Execution Behavior

Python reads files from top to bottom. If the same method name is defined multiple times inside a class, the last definition overwrites all previous ones. This happens due to name binding, not because Python follows a bottom-up approach. Python does not support method overloading by default.

4. Phone Number Field: Design and Validation

Phone numbers are stored using `CharField` instead of `IntegerField` because they are identifiers, not numeric values. They may start with zero or include a country code such as +91. Using `CharField` alone allows invalid input like alphabets, so Django validators such as `RegexValidator` are used to enforce digit-only input and proper length validation.

Interview-Ready Summary

Django choices separate database values from display labels. `__str__()` executes only when a string representation is required. The last method definition overrides earlier ones in Python. Phone numbers should be validated using `CharField` with `RegexValidator` to ensure correctness.